

Transform Mobility

CLINICAL APPLICATIONS

- Maintains Healthy Joint Function for Normal Mobility and Lubrication
- Provides Precursors for Generation of Cartilage, Tendons, Ligaments, Fascia, and Skin
- Promotes a Healthy Joint Environment by Supporting Normal Activity of Cytokines and Catabolic Enzymes



OVERVIEW

Collagen Factors provide targeted precursors and cofactors known to be involved in the biochemical processes that support healthy connective tissue in the skin, tendons, ligaments, cartilage, joints, and bone extracellular matrix. This formulation contains patented hyaluronic acid, MSM, free-form amino acids, key minerals, and vitamin C.

Suggested Use:

2 capsules per day or as recommended by your health care professional.

INGREDIENT BENEFITS

Hyaluronic Acid Extract

Mobilee® is a patented, high-molecular-weight hyaluronic acid (HA), which enhances the viscoelastic, lubricating properties of synovial fluid. In addition, it contains collagen and other glycosaminoglycans (GAGs) to enhance skin and joint health. Hyaluronic acid also plays a role in the biophysical, biochemical, and cell regulation processes in joint synovial tissues. Scientific evidence shows that Mobilee® supports chondrocytes and synovial cells function and is two to four times more active than fermented sources of HA in nourishing and supporting the health of synovial fluid. The latest clinical research includes HA in proactive and maintenance approaches to joint care that supports normal range-of-motion. Research suggests that Mobilee® supports the quality of synovial fluid by positively influencing synovial HA concentration, and by reducing the expression of degradative factors in synovial fluid.

Specific Precursors

Collagen is comprised of and formed from the amino acids proline and lysine. Lysine is an essential amino acid the body relies on adequate dietary intake for its involvement in the synthesis of connective tissues. Proline is well-known for its involvement in the triple helix of collagen—it is regularly spaced with glycine in the amino acid sequence of collagen fiber, which together comprises about 50% of total collagen sequencing. Proline contributes to not only collagen formation but muscle, tendons, and skin. Lysine helps support the linking and stabilization of collagen and contributes to the collagen matrix formation of veins, arteries, and capillaries. During procollagen synthesis, both proline and lysine are hydroxylated to hydroxyproline and hydroxylysine, a reaction requiring vitamin C as a cofactor. Inadequate

levels of vitamin C can impact this reaction, resulting in the incomplete formation of the helical structure, which leads to a delicate and easily destroyed structure. Vitamin C is combined with two amino acids—lysine and proline—to form procollagen. Procollagen is then used to manufacture one of several types of collagen found in different tissues throughout the body.

Targeted Cofactors

Zinc is an essential trace mineral that supports DNA synthesis, cell division, cell membrane structure, immune function, and protein synthesis. Requirements are highest when the body's repair system is most active. MSM is responsible for the flexible disulfide bonds between cells, including those that make up connective tissue and the skin. It blocks undesirable chemical and physical cross-linking of collagen and preserves connective tissue pliancy, which is crucial for supporting healthy collagen. MSM has been shown to significantly increase plasma glutathione levels, an additional benefit in maintaining collagen's defensive mechanisms against oxidative stress. Multiple clinical studies also show that MSM, in combination with other joint health ingredients, supports tendon and joint health. Manganese assists the growth and development of normal bone and the synthesis of cartilage. Manganese is also a structural component of antioxidant enzymes, and aids in amino acid production. During the tissue synthesis process, which requires increased production of collagen, manganese is required for the activation of prolidase, an enzyme that functions to provide the amino acid proline, for collagen formation in human skin cells. Nutritional co-factors required for tissue regeneration include zinc, vitamin C, and amino acids. Inadequacy of these nutrients impacts endogenous collagen production and connective tissue synthesis.

Collagen Factors



CLINICAL APPLICATIONS

- *Maintains Healthy Joint Function for Normal Mobility and Lubrication*
- *Provides Precursors for Generation of Cartilage, Tendons, Ligaments, Fascia and Skin*
- *Promotes a Healthy Joint Environment by Supporting Normal Activity of Cytokines and Catabolic Enzymes*

MUSCULOSKELETAL HEALTH

Collagen Factors provides targeted precursors and cofactors known to be involved in the biochemical processes that support healthy connective tissue in the skin, tendons, ligaments, cartilage, joints, and bone extracellular matrix. This formulation contains patented hyaluronic acid, MSM, free-form amino acids, key minerals and vitamin C.

Hyaluronic Acid Extract[†]

Mobilee® is a patented, high-molecular-weight hyaluronic acid (HA), which enhances the viscoelastic, lubricating properties of synovial fluid. In addition, it contains collagen and other glycosaminoglycans (GAGs) to enhance skin and joint health. Hyaluronic acid also plays a role in the biophysical, biochemical and cell regulation processes in joint synovial tissues. Scientific evidence shows that Mobilee® supports chondrocytes and synovial cells function, and is two to four times more active than fermented sources of HA in nourishing and supporting the health of synovial fluid.^{1,2} The latest clinical research includes HA in proactive and maintenance approaches to joint care that supports normal range-of-motion. Research suggests that Mobilee® supports the quality of synovial fluid by positively influencing synovial HA concentration, and by reducing the expression of degradative factors in synovial fluid.³⁻⁶

Specific Precursors[†]

Collagen is comprised of and formed from the amino acids proline and lysine. Lysine is an essential amino acid the body relies on adequate dietary intake for its involvement in the synthesis of connective tissues. Proline is well-known for its involvement in the triple helix of collagen—it is regularly

spaced with glycine in the amino acid sequence of collagen fiber, which together comprise about 50% of total collagen sequencing. Proline contributes to not only collagen formation, but muscle, tendons and skin. Lysine helps support the linking and stabilization of collagen, and contributes to the collagen matrix formation of veins, arteries and capillaries. During procollagen synthesis, both proline and lysine are hydroxylated to hydroxyproline and hydroxylysine, a reaction requiring vitamin C as a cofactor.⁹⁻¹¹ Inadequate levels of vitamin C can impact this reaction, resulting in incomplete formation of the helical structure, in which leads to a delicate and easily destroyed structure. Vitamin C is combined with two amino acids—lysine and proline—to form procollagen. Procollagen is then used to manufacture one of several types of collagen found in different tissues throughout the body.¹²⁻¹³ Furthermore, an animal study suggests that serum ascorbate levels influence fluid accumulation in the joint, reinforcing the importance of this cofactor in joint health.¹²

Targeted Cofactors

Zinc is an essential trace mineral that supports DNA synthesis, cell division, cell membrane structure, immune function, and protein synthesis. Requirements are highest when the body's repair system is most active. **MSM** is responsible for the flexible disulfide bonds between cells, including those that make up connective tissue and the skin. It blocks undesirable chemical and physical cross-linking of collagen and preserves connective tissue pliancy, which is crucial for supporting healthy collagen. MSM has been shown to significantly increase plasma glutathione levels, an additional benefit in maintaining collagen's defensive mechanisms against oxidative stress.

[†] These statements have not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease.

Multiple clinical studies also show that MSM, in combination with other joint health ingredients, supports tendon and joint health.⁶⁻⁷ **Manganese** assists the growth and development of normal bone and the synthesis of cartilage. Manganese is also a structural component of antioxidant enzymes, and aids in amino acid production. During the tissue synthesis process, which requires an increased production of collagen, manganese is required for the activation of prolydase, an enzyme that functions to provide the amino acid proline, for collagen formation in human skin cells.¹⁵ Nutritional co-factors required for tissue regeneration include zinc, vitamin C and amino acids. Inadequacy of these nutrients impacts endogenous collagen production and connective tissue synthesis.

Directions

2 capsules per day or as recommended by your health care professional.

Does Not Contain

Gluten, yeast, artificial colors or flavors.

Cautions

If you are pregnant or nursing, consult your physician before taking this product.

Supplement Facts ^{V1}		
Serving Size 2 Capsules		
Servings Per Container 30		
	Amount Per Serving	% Daily Value
Vitamin C (as Ascorbic Acid USP)	100 mg	111%
Zinc (as TRAACS™ Zinc Bisglycinate Chelate)	15 mg	136%
Manganese (as TRAACS™ Manganese Bisglycinate Chelate)	10 mg	435%
Methylsulfonylmethane (MSM) (OptiMSM®)	500 mg	*
L-Lysine Hydrochloride USP	200 mg	*
L-Proline USP	200 mg	*
Mobilee® (Standardized to contain 40 mg Hyaluronic Acid)	80 mg	*
* Daily Value not established.		

ID# 332060 60 Capsules

References

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